Physics — Topics to Master

From the baseline diagnostic (Grade 7–9 program, 28 Jun 2026). Tick each box when he can explain it in English and solve a simple problem on it.

61 topics to master — ● **46** not yet known ● **15** shaky. Grouped by topic, largest first.

Forces & mechanics (17 to master)

Read: NCERT ch4 & ch6 · Cambridge Stage 8 §3

- [ ] ● Describe the effect of balanced and unbalanced forces on motion. g8-6
- [ ] ● Describe friction, and give an example where it is useful and where it is a nuisance. g8-7
- [ ] ● Find the turning effect of a force using moment = force × distance. g8-8
- [ ] ● Explain the difference between speed and velocity. g8-4
- [ ] ● What does acceleration mean? g8-5
- [ ] ● Name several different types of force (gravity, friction, magnetic, contact). g7-4
- [ ] ● Describe gravity as a force of attraction between masses. g7-6
- [ ] ● Define velocity, and explain how it differs from speed. g9-1
- [ ] ● Define acceleration and state its unit. g9-2
- [ ] ● Interpret a speed–time graph: what do the gradient and the area underneath represent? g9-3
- [ ] ● State Newton's first law (inertia) with an example. g9-4
- [ ] ● Use Newton's second law, F = m × a, in a calculation. g9-5
- [ ] ● State Newton's third law with an example. g9-6
- [ ] ● Define momentum (p = m × v) and calculate it. g9-7
- [ ] ● State the idea of conservation of momentum (e.g. in a collision). g9-8
- [ ] ● Describe gravitation as a universal attraction between masses. g9-9
- [ ] ● What is free fall, and what is the approximate value of g near the Earth's surface? g9-10

Electricity & magnetism (12 to master)

Read: Siyavula ch15–17 · CK-12 ch23–25 · Cambridge Stage 7/8/9 §9

- [ ] ● Describe electric current as a flow of charge/electrons. g7-16
- [ ] ● Explain the difference between a conductor and an insulator, with an example of each. g7-17
- [ ] ● Build a simple series circuit and draw the symbols for a cell, switch, lamp and ammeter. g7-18
- [ ] ● Describe a magnet and its magnetic field, and say which materials are magnetic. g8-24
- [ ] ● How do you make an electromagnet, how can you make it stronger, and give a use. g8-25
- [ ] ● Define electric current and state its unit. g9-17
- [ ] ● Define potential difference (voltage) and state its unit; how is a voltmeter connected? g9-18
- [ ] ● Define resistance and state its unit. g9-19
- [ ] ● State and use Ohm's law (V = I × R). g9-20
- [ ] ● Compare how current and voltage behave in series versus parallel circuits. g9-21
- [ ] ● Calculate electrical power (P = V × I). g9-22
- [ ] ● Describe that an electric current produces a magnetic effect (electromagnetism), and give a use. g9-23

Energy & work (incl. heat) (11 to master)

Read: NCERT ch7 · CK-12 ch17–18 · Cambridge Stage 7 §3.5–3.7

- [ ] Name several forms/stores of energy and describe one energy transfer. g7-8
- [ ] State that energy is conserved, and that it tends to dissipate (spread out, become less useful). g7-9
- [ ] Explain the difference between heat and temperature. g8-12
- [ ] Describe thermal expansion of solids, liquids and gases, with an everyday example. g8-14
- [ ] Describe conduction and give an example. g8-15
- [ ] Describe radiation (infrared) and give an example. g8-17
- [ ] Calculate work done (work = force × distance) and state its unit. g9-11
- [ ] Calculate kinetic energy (½ × m × v²). g9-12
- [ ] Calculate the change in gravitational potential energy (m × g × h). g9-13
- [ ] Define power (work / time) and calculate it. g9-14
- [ ] What is efficiency, and how is it calculated? g9-15

Optics / light (6 to master)

Read: Cambridge Stage 8 §6.1–6.3 · CK-12 ch22

- [ ] State that light travels in straight lines, and explain how we are able to see objects. g8-18
- [ ] State and use the law of reflection. g8-19
- [ ] Describe the image formed in a plane mirror (upright, same size, laterally inverted, virtual). g8-20
- [ ] Describe refraction — light bending when it passes into glass or water — and why it happens. g8-21
- [ ] What does a prism do to white light? g8-22
- [ ] Why does a coloured object (e.g. a red apple) look that colour? g8-23

Waves & sound (6 to master)

Read: NCERT ch10 · Siyavula ch8–10 · Cambridge Stage 9 §6

- [ ] What causes an echo? g7-14
- [ ] What changes the pitch, and what changes the loudness, of a sound? g7-15
- [ ] Define the amplitude, wavelength, frequency and period of a wave. g9-24
- [ ] Explain the difference between transverse and longitudinal waves, with an example of each. g9-25
- [ ] Use the wave equation: wave speed = frequency × wavelength. g9-26
- [ ] Describe the properties of sound waves (reflection/echo, pitch ↔ frequency, loudness ↔ amplitude). g9-27

Pressure & fluids (4 to master)

Read: CK-12 ch15 · Cambridge Stage 8 §3.5–3.6

- [ ] Define pressure (force / area); explain why skis or wide tyres reduce pressure on the ground. g8-9
- [ ] Use the particle model to explain pressure in liquids and gases, and how it changes with depth. g8-10
- [ ] What is atmospheric pressure, and give one piece of evidence that it exists. g8-11
- [ ] Explain upthrust (buoyancy), and relate floating to density and upthrust. g9-16

The atom (2 to master)

Read: Siyavula ch4 · CK-12 ch5

- [ ] Describe the structure of the atom (a nucleus of protons and neutrons, with electrons around it). g9-28
- [ ] What do atomic (proton) number and mass (nucleon) number mean? g9-29

Maths & practical skills (3 to master)

Read: Practice via the test's problems and worked examples

- [ ] Rearrange a simple formula: from speed = distance / time, make distance the subject. sk-1
- [ ] Substitute numbers into a formula and give the answer with the correct unit. sk-2
- [ ] Read a line graph: what does the gradient (steepness) tell you, and what does a flat line mean? sk-4